DIGITAL IMAGE PROCESSING · LECTURE 1

# Image Acquisition & Intensity Transformations

*From physical signal to digital image — and the first tools used to enhance it*

Digital Image Fundamentals · Signal Processing

ROADMAP

# Today's Agenda

PART I

## Image Acquisition

1. **Digital cameras**
   Bayer colour filter array & demosaicing
2. **MRI**
   k-space and the inverse FFT
3. **CT**
   Radon transform & the sinogram
4. **Ultrasound**
   Pulse–echo B-mode imaging

PART II

## Intensity Transformations

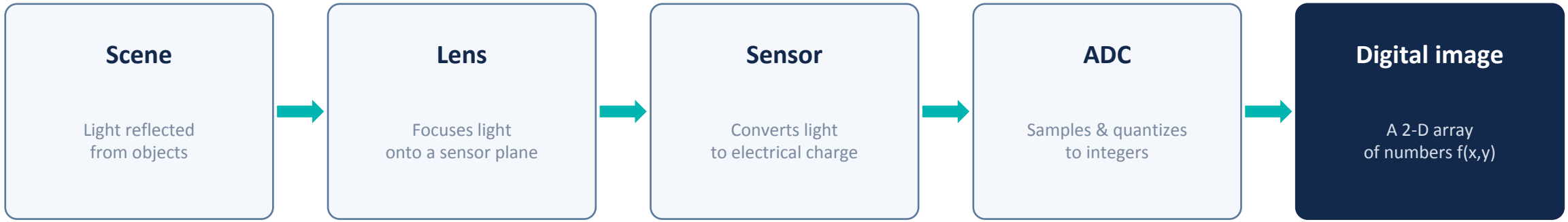
1. **Point processing**
   The model s = T(r)
2. **Negative, log, power-law**
   Basic transformation functions
3. **Piecewise-linear functions**
   Contrast stretching & intensity slicing
4. **Bit-plane slicing**
   Decomposing an image by bit significance

PART I

# Image Acquisition

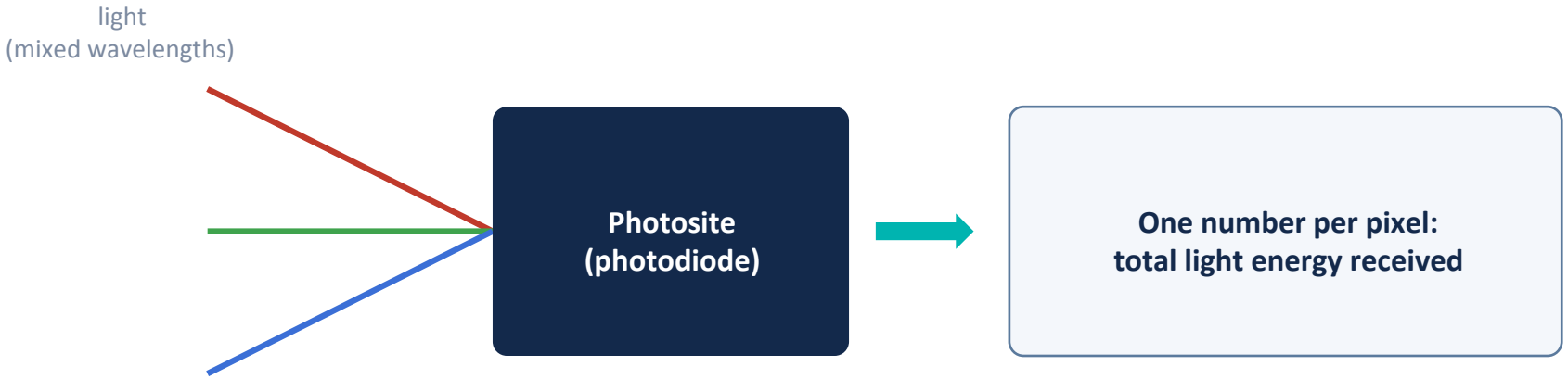
*How a physical signal becomes a digital image*

# From Scene to Digital Image




Every stage discards or transforms information — acquisition design determines what is even possible to recover later.

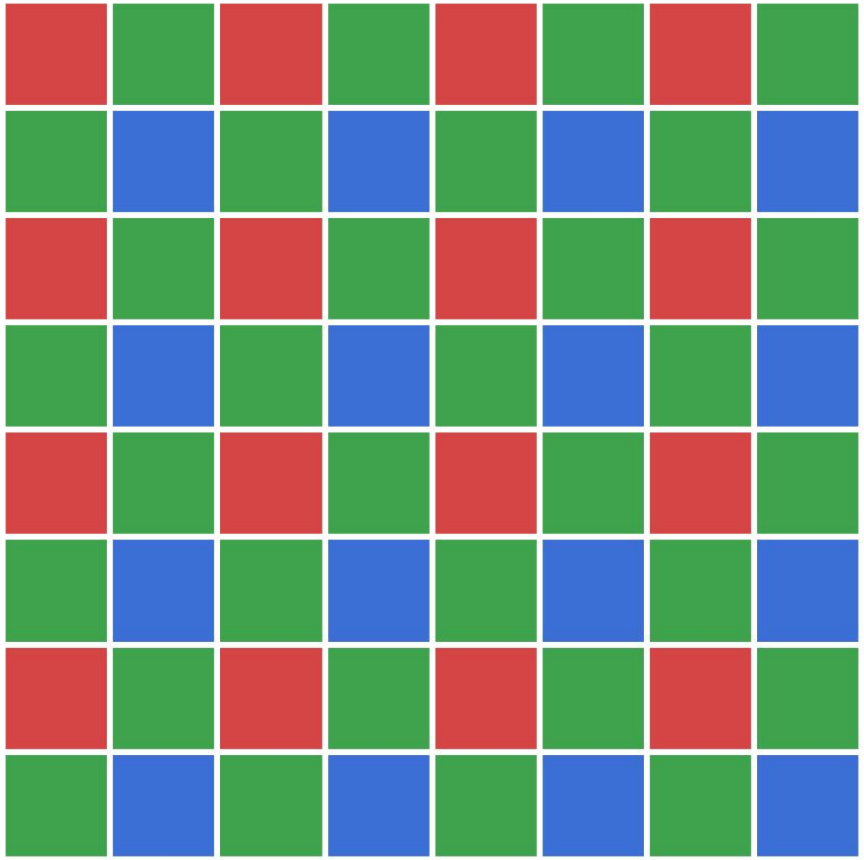
# A Sensor, by Itself, Cannot See Colour




*A photosite integrates all wavelengths together — brightness only, no colour identity.*

**Fix: place a coloured filter over every photosite, so each one measures only R, G, or B.**

# The Bayer Colour Filter Array



A filter mosaic sits directly over the sensor: one colour per photosite.

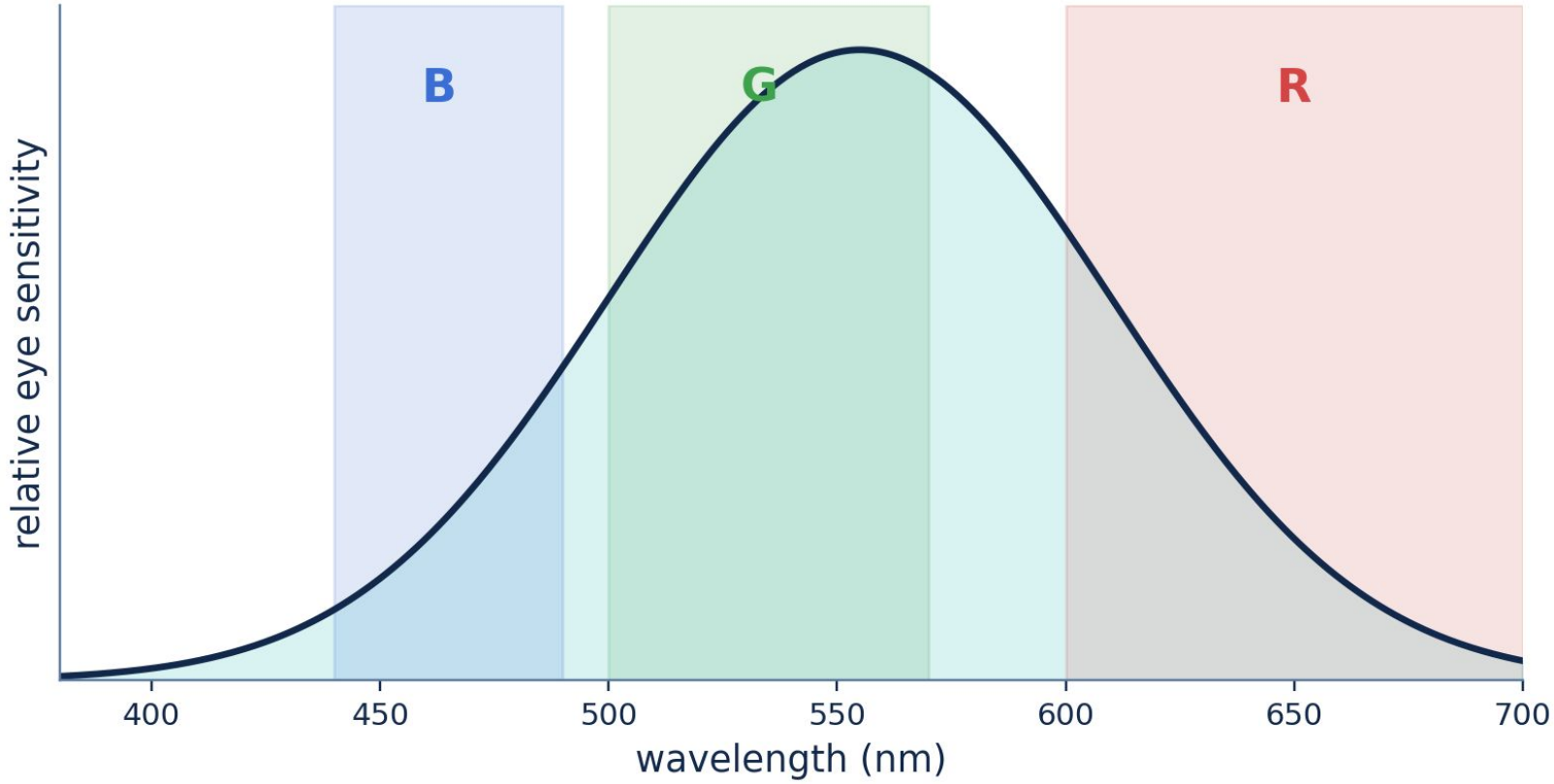
**Red**
25% of photosites

**Green**
50% of photosites

**Blue**
25% of photosites

*Named for Bryce Bayer (Kodak, 1976). The 2×2 repeating unit is R,G / G,B — called the RGGB pattern.*

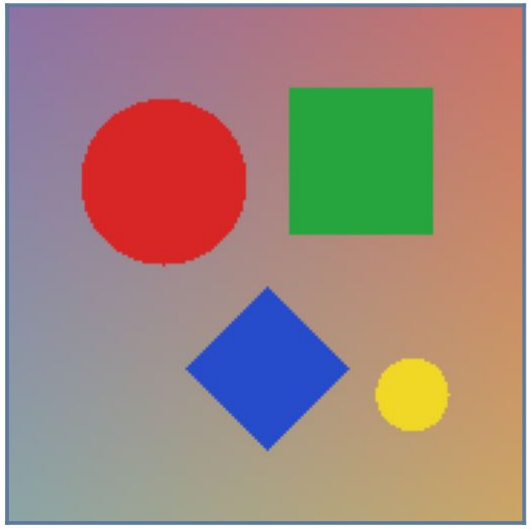
# Why More Green?



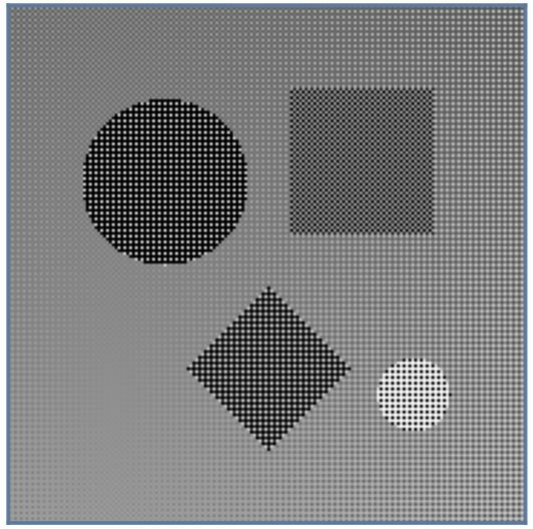

The human eye is most sensitive near 555 nm (green). Weighting the CFA toward green better matches perceived sharpness and reduces visible noise.

# Demosaicing: From Mosaic to Full Colour

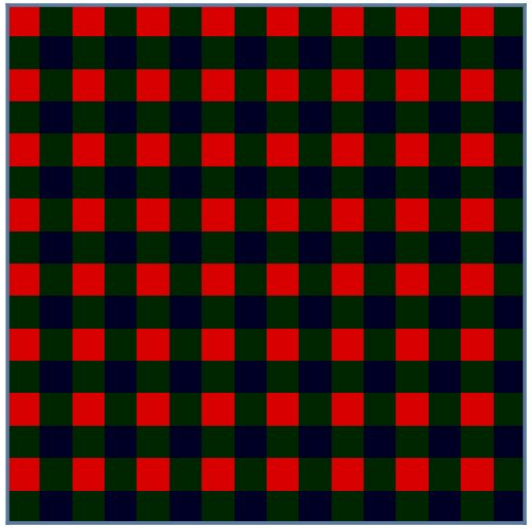
Scene



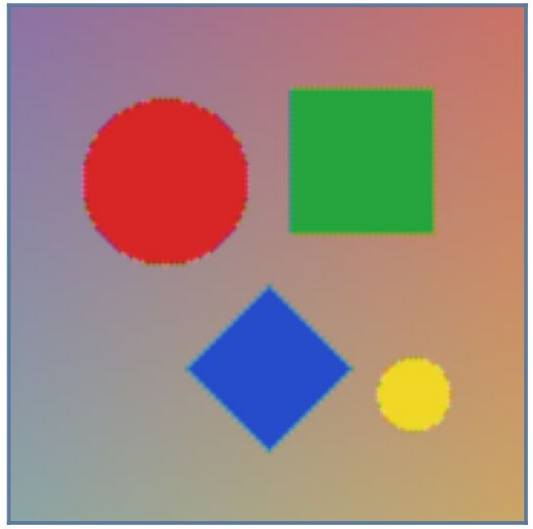
Raw sensor read-out
(one channel per pixel)



Zoom: CFA pattern
(each cell tags a colour)

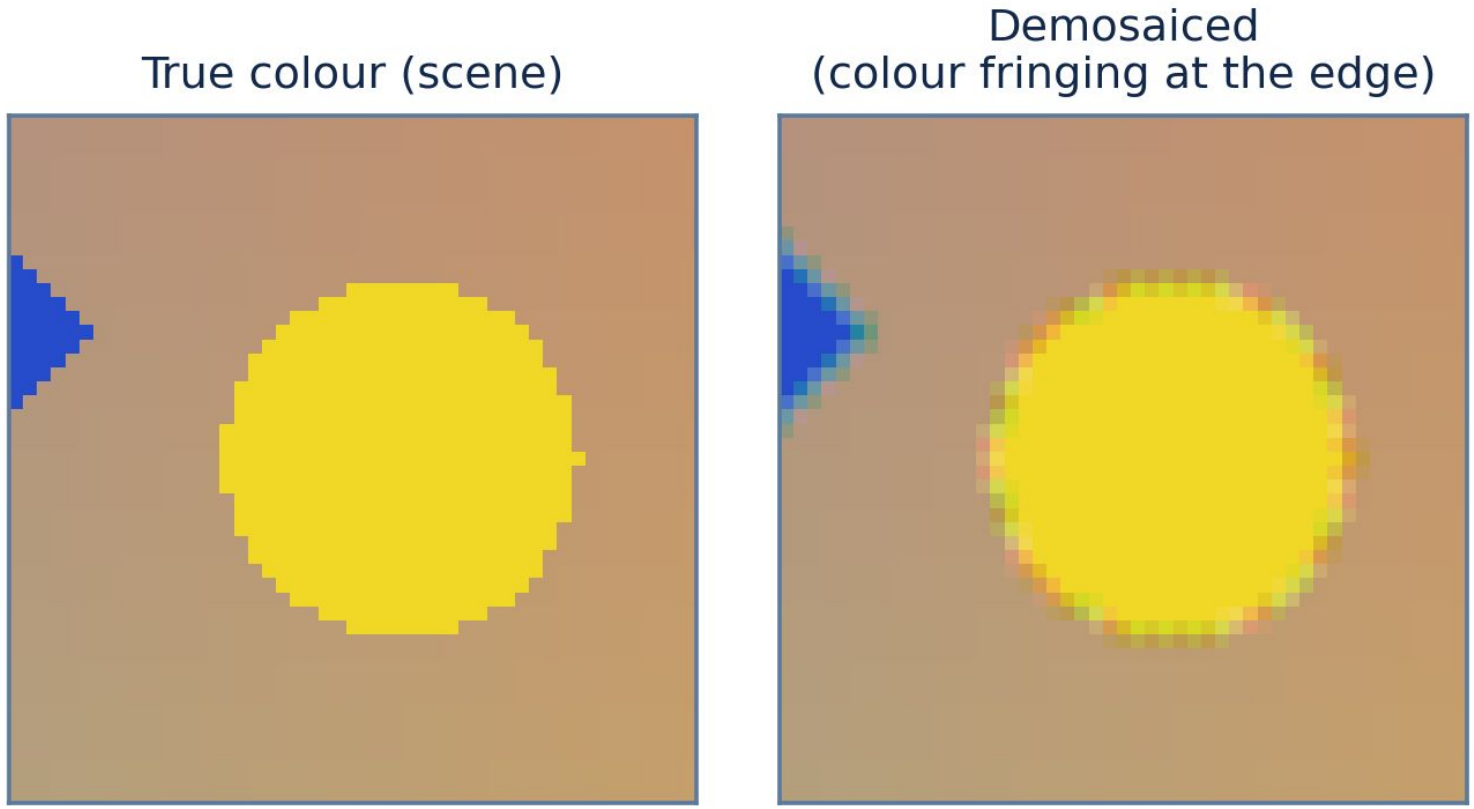


Demosaiced
(missing colours interpolated)



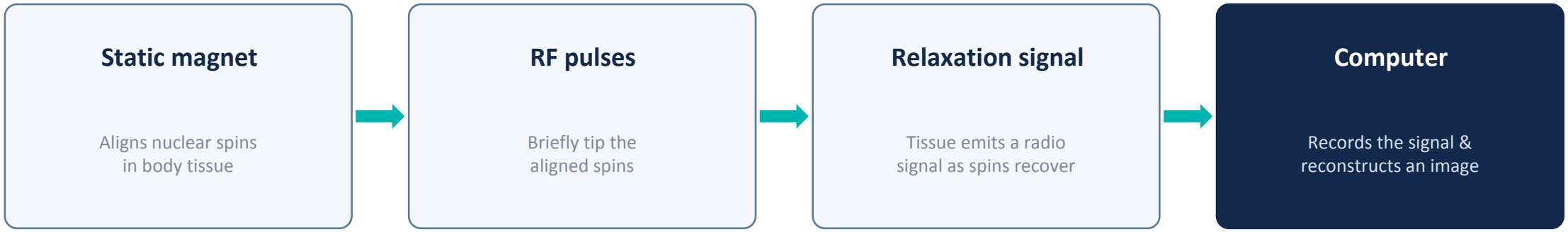
Each pixel is missing two of its three colours. Demosaicing interpolates the missing values from neighbouring photosites of the same colour.

# Demosaicing Artifacts




Interpolation is a guess. At sharp colour edges, neighbouring photosites disagree — producing colour fringing ("zippering") along high-frequency boundaries.

# MRI: the Basic Idea (Not the Physics)




No ionizing radiation. Signal strength and location are computed, not photographed directly — the image is reconstructed.

# What the Scanner Actually Measures

Each RF pulse and magnetic-field gradient does not record a pixel. It records one Fourier coefficient of the image.

**Raw scanner signal**

sampled over time, one frequency at a time



**k-space F(u, v)**

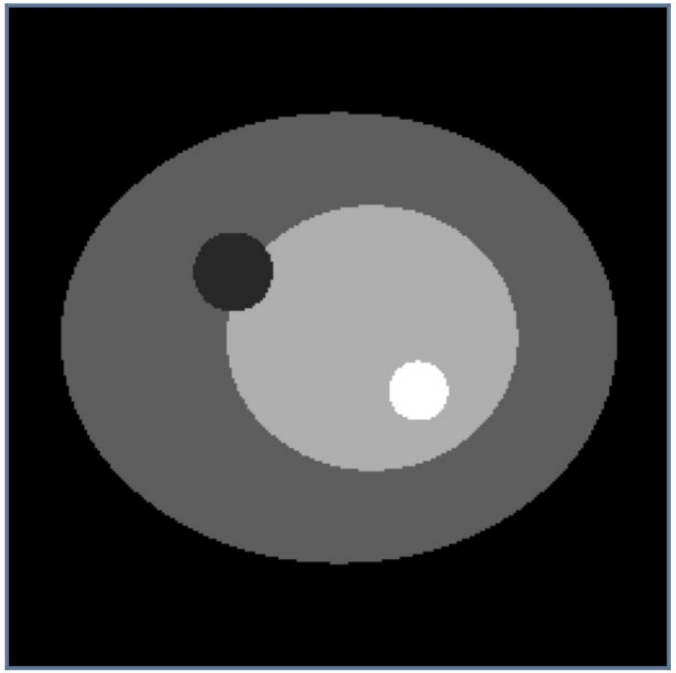
the spatial-frequency domain — not the image itself

$$\text{Image } f(x, y) = \text{IFFT}[F(u, v)]$$

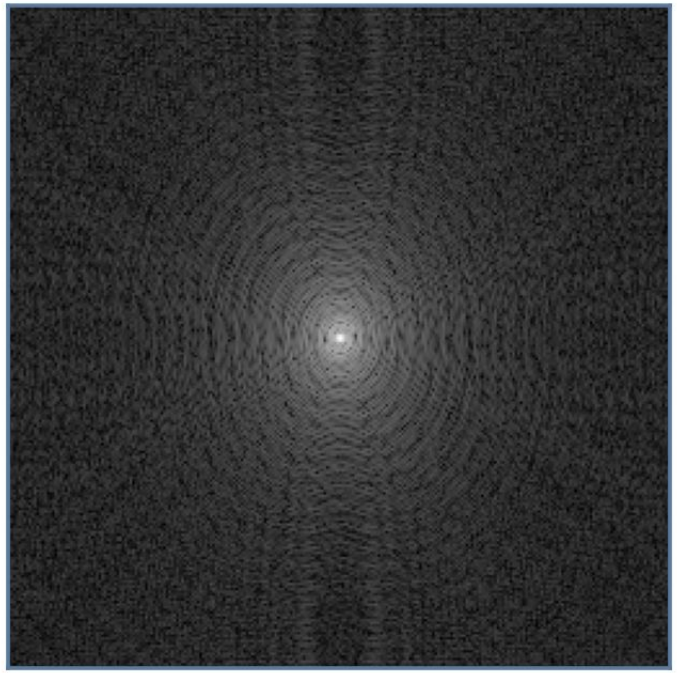
MRI reconstruction is, at its core, a 2-D inverse Fourier transform problem.

# Object ⇄ k-Space ⇄ Reconstruction

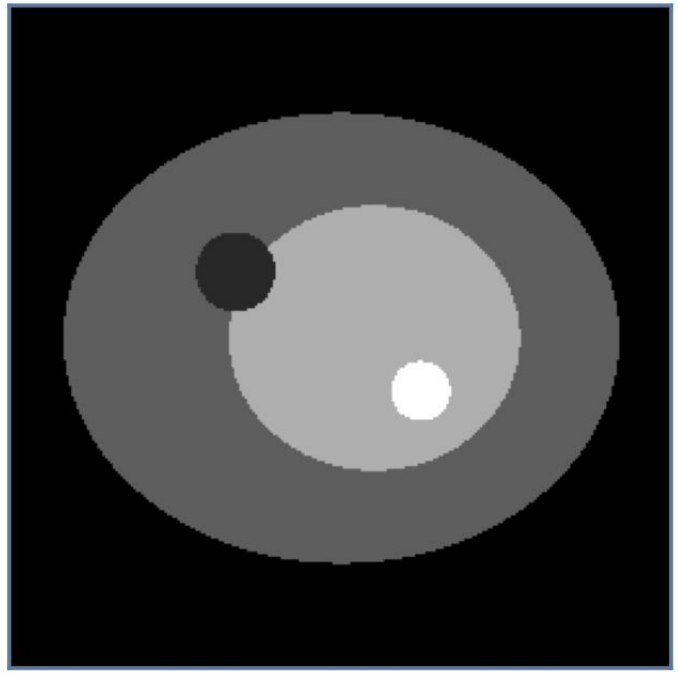
Object
$f(x, y)$



k-space (log magnitude)
$F(u, v)$

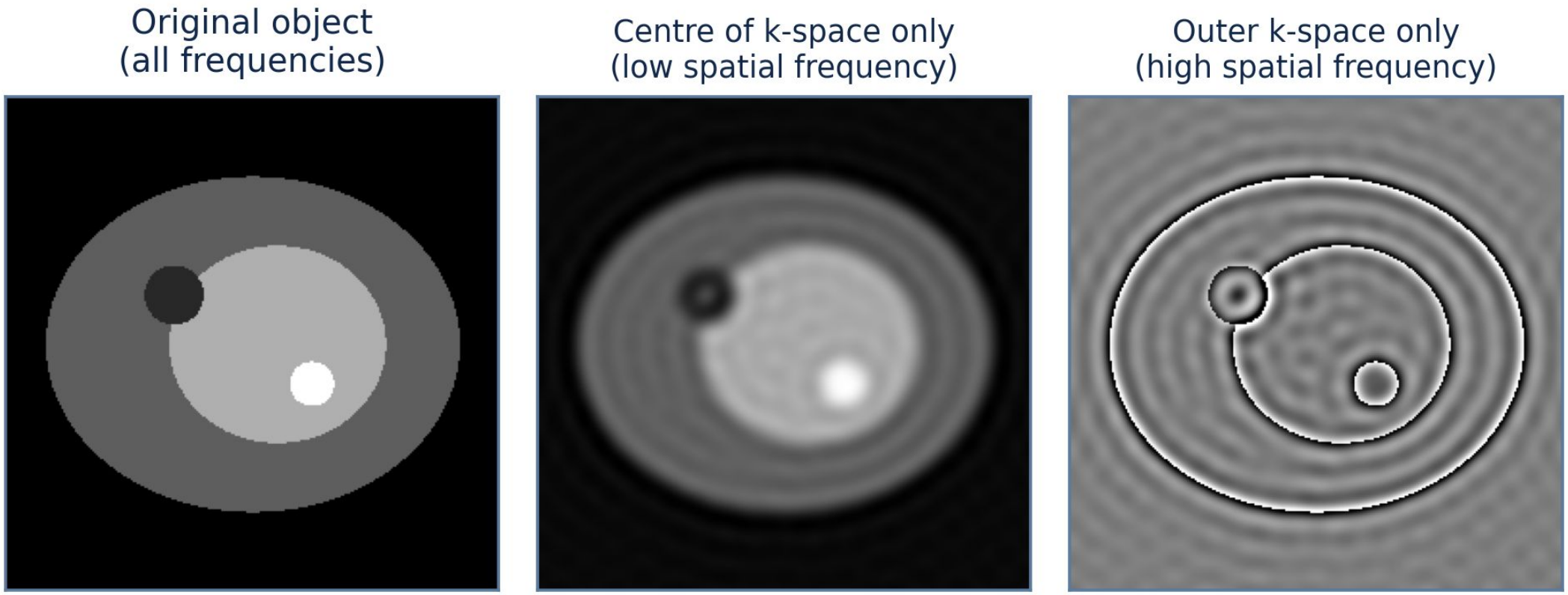


Reconstruction
IFFT[$F(u, v)$]



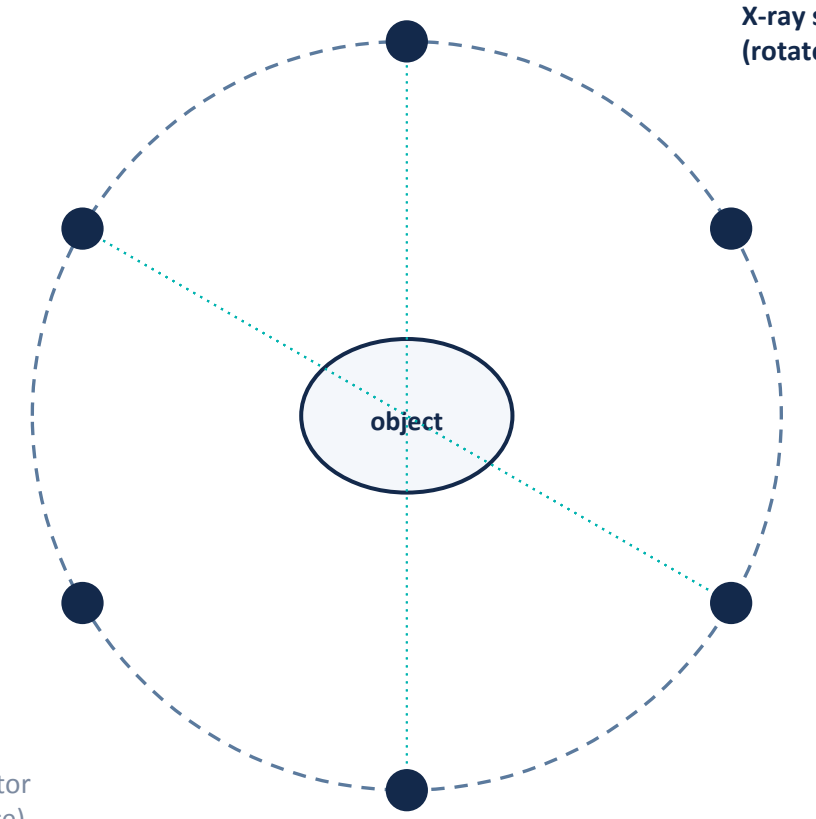
A synthetic test object, its 2-D Fourier transform (log magnitude), and the exact reconstruction obtained by inverse-transforming it back.

# k-Space Structure: Centre vs Edges




Keeping only the centre of k-space preserves contrast and gross shape; keeping only the outer region preserves edges and fine detail.

# CT: a Rotating X-ray Source and Detector

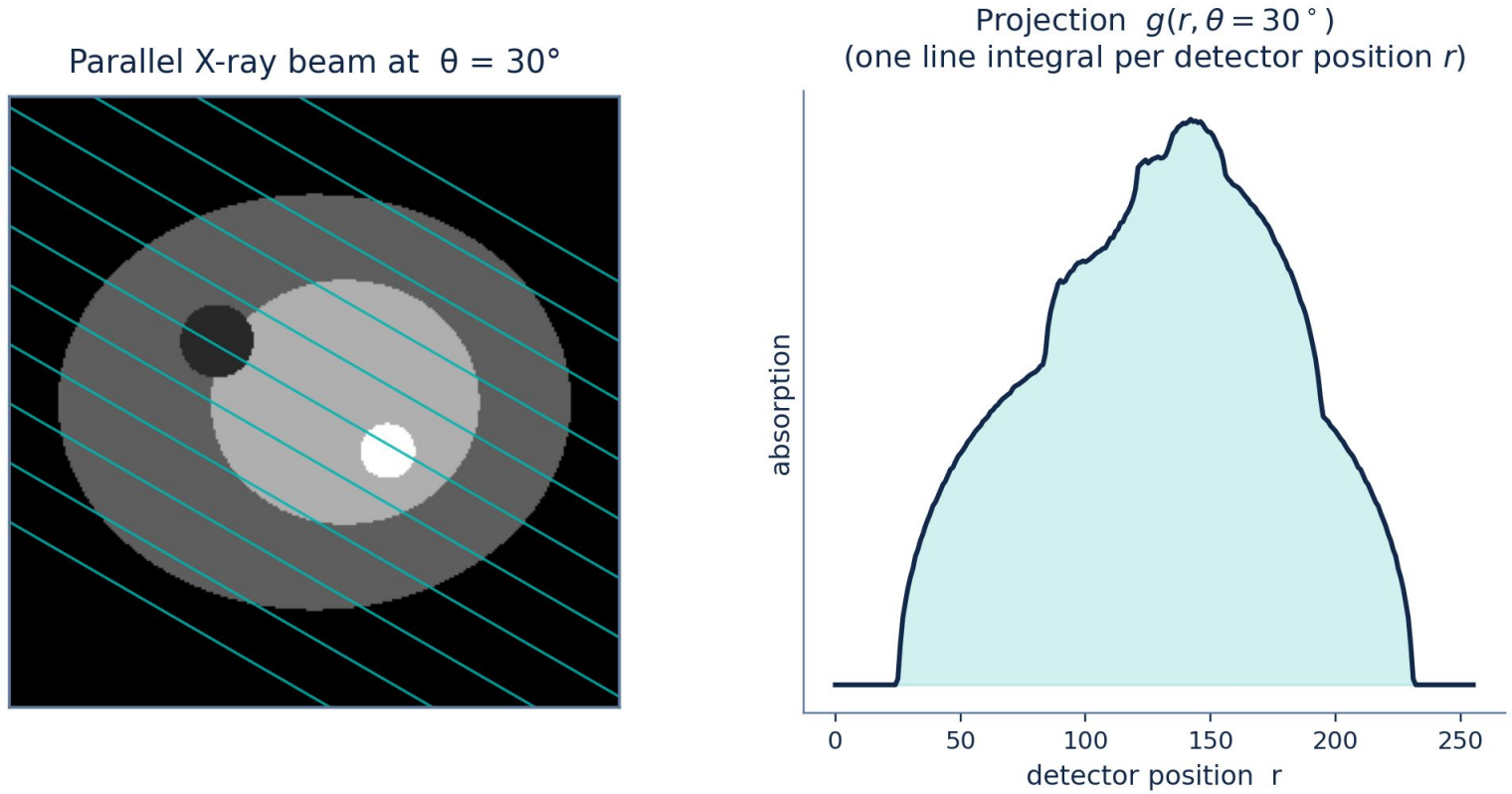



For every angle θ, the detector measures how much X-ray energy passed through the object.

Each measurement is a projection — a 1-D summary of the object along that direction.

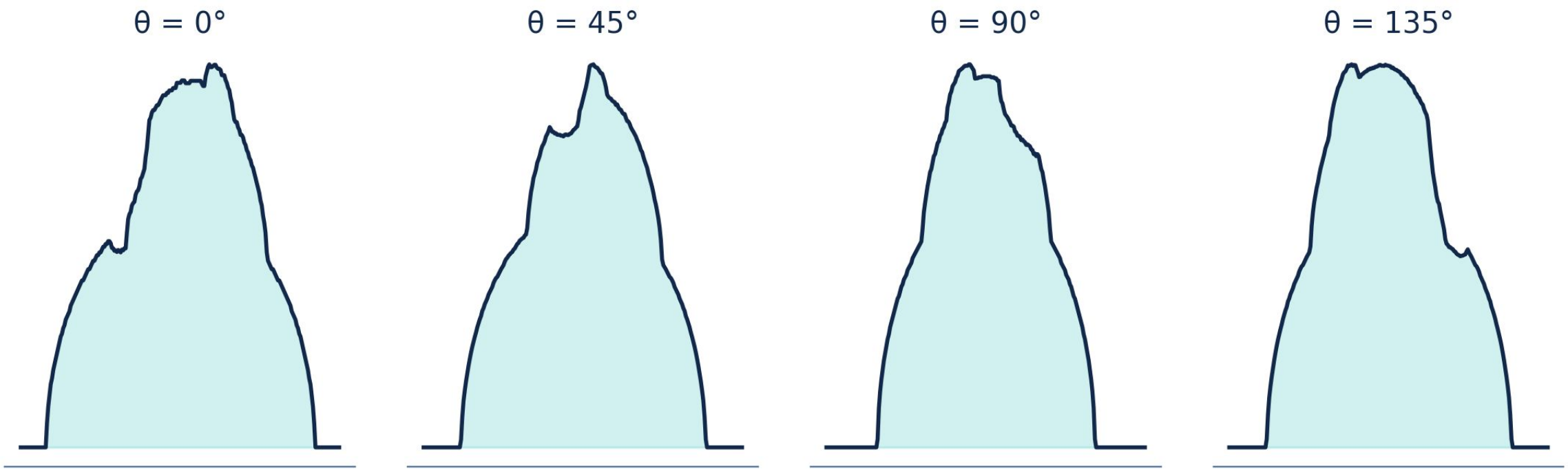
Repeating this across 0°–180° gives a complete set of projections, enough to reconstruct the full cross-section.

# The Radon Transform: Projection = Line Integral



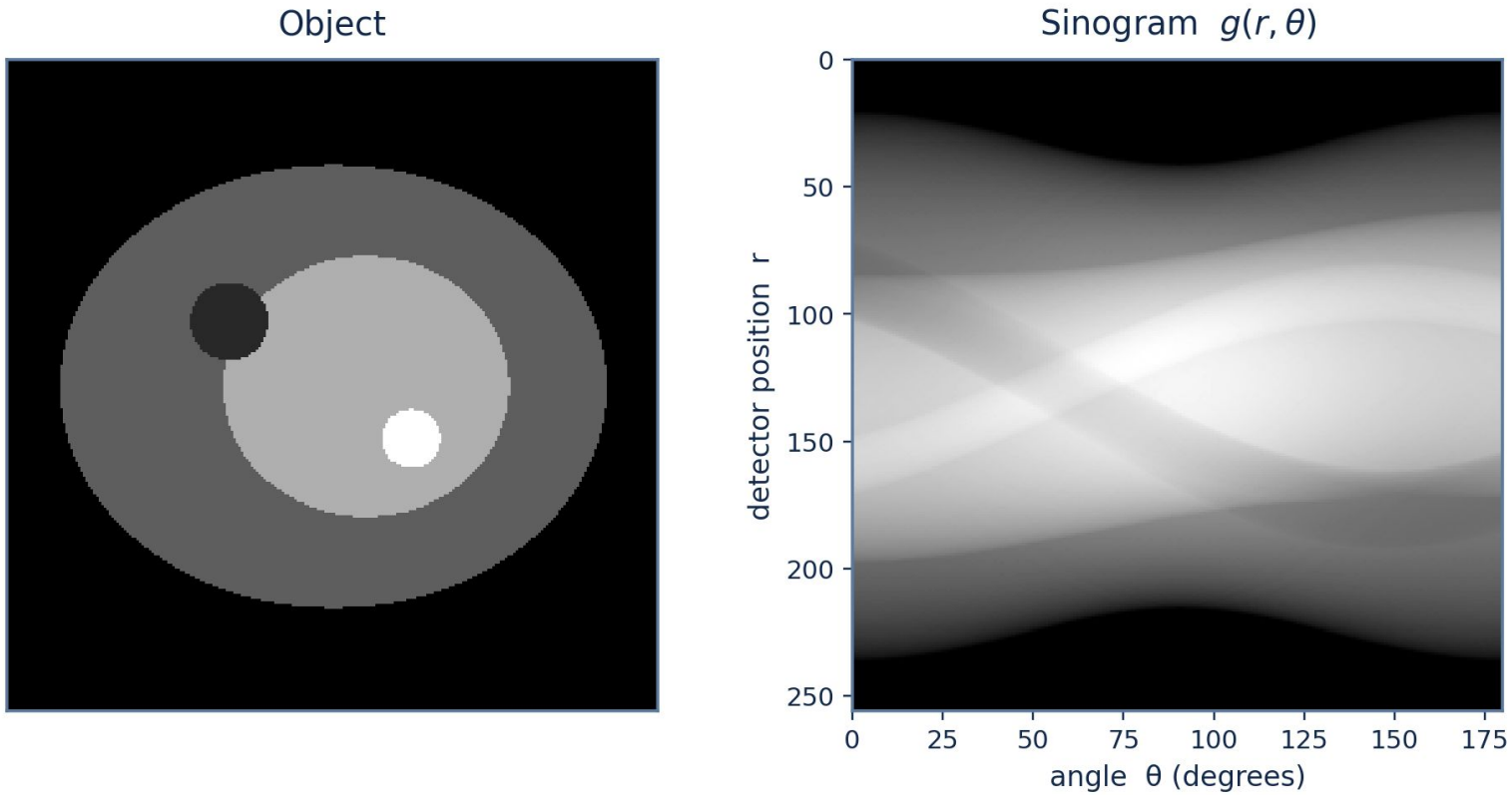

A parallel-ray beam at angle θ. Each detector position r records the line integral of absorption along its ray: this is the Radon transform g(r, θ).

# Many Angles, Many Projections



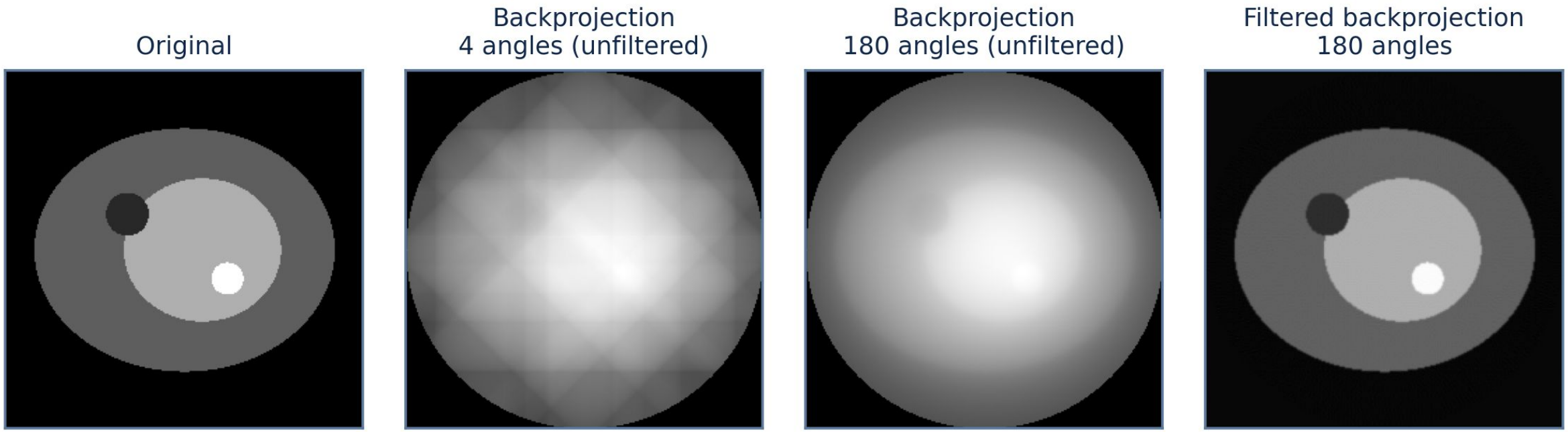

The projection profile changes shape as the source/detector pair rotates — each angle sees the object differently.

# The Sinogram



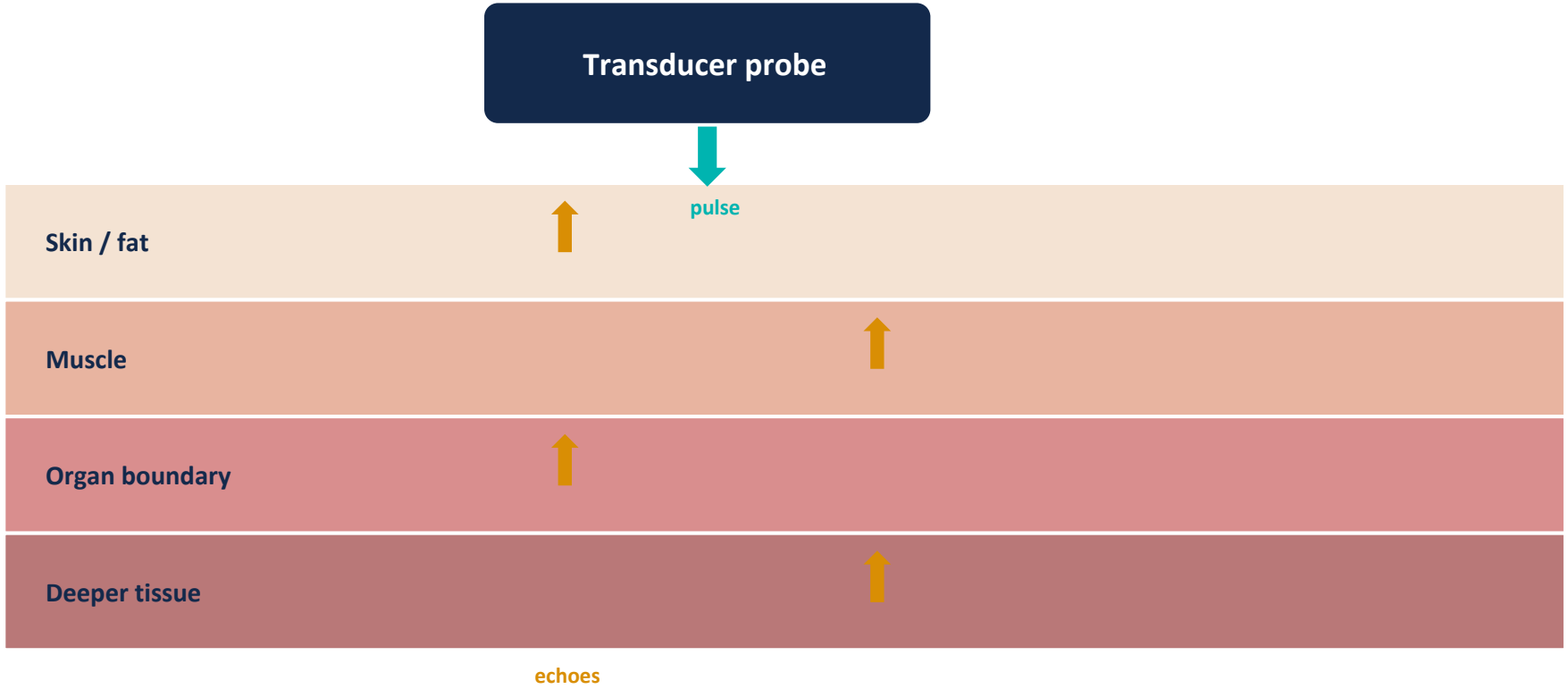

Stacking every projection g(r, θ) as a row, indexed by angle, forms the sinogram — a complete, invertible representation of the object.

# From Sinogram to Image: Backprojection



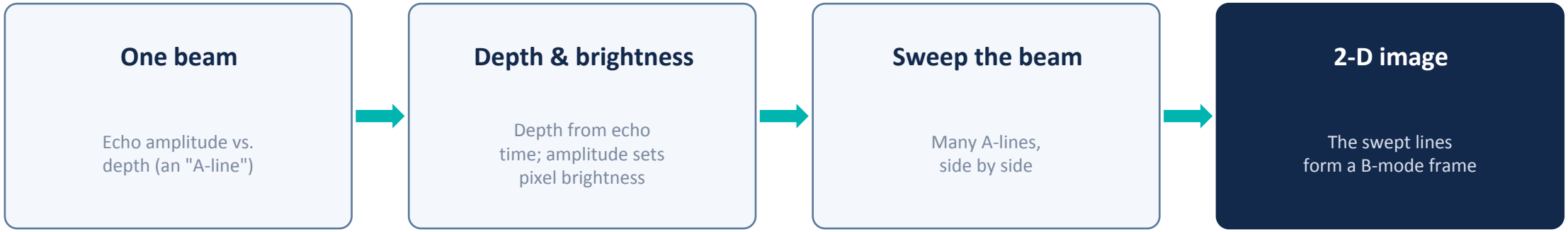

Simple backprojection smears each ray back across the image; more angles sharpen it but leave residual blurring — filtering (filtered backprojection) removes it.

# Ultrasound: Pulse–Echo Imaging




Short pulses (1–5 MHz) partially reflect at every tissue boundary; the rest continues deeper until the next boundary.

# From Echoes to a B-Mode Image




Depth is computed from echo time and the speed of sound in tissue (≈1540 m/s); this repeats thousands of times per second.

# Characteristics of Ultrasound Imaging

### Real-time

Frames form fast enough to watch live motion

### No ionizing radiation

Safe for repeated use, including in pregnancy

### Portable & low-cost

Compact systems, no special shielded room

### Operator-dependent

Image quality relies heavily on probe skill

### Blocked by bone & air

Sound barely penetrates these boundaries

### Doppler capability

Can measure blood-flow direction and speed

PART II

# Intensity Transformations

*The simplest form of image processing: reshaping intensity, pixel by pixel*

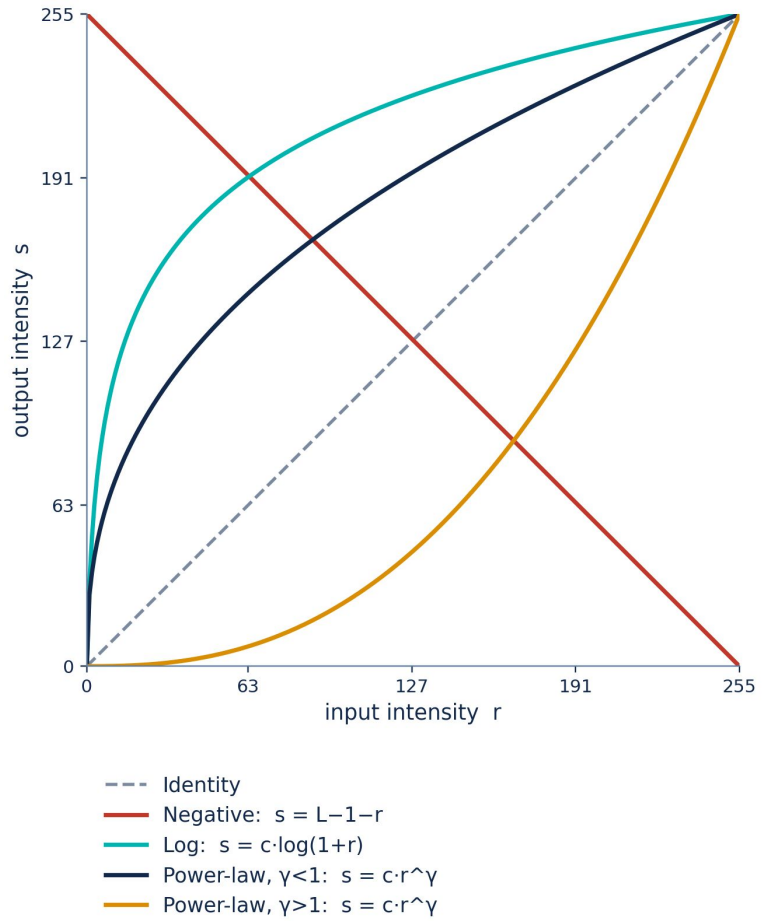
# Point Processing: s = T(r)




**Output at a pixel depends only on the input value at that same pixel — never on its neighbours.**

*This is what separates intensity transformations from spatial filtering, which we cover later.*

# A Family of Basic Transformation Functions




**Negative**

Reverses the intensity scale entirely

**Log**

Expands dark values, compresses bright ones

**Power-law, γ<1**

Same shape as log, but tunable

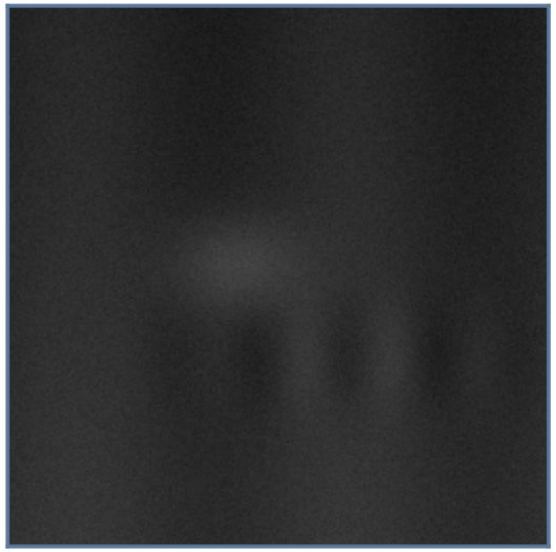
**Power-law, γ>1**

Expands bright values, compresses dark ones

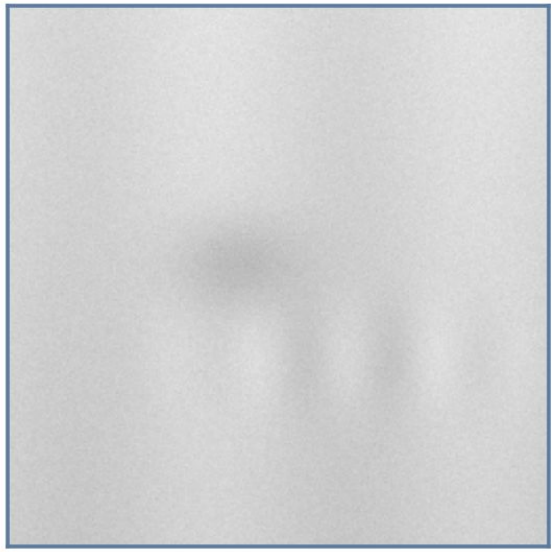
*All are special cases of reshaping the mapping s = T(r) between input and output intensity.*

# Image Negative

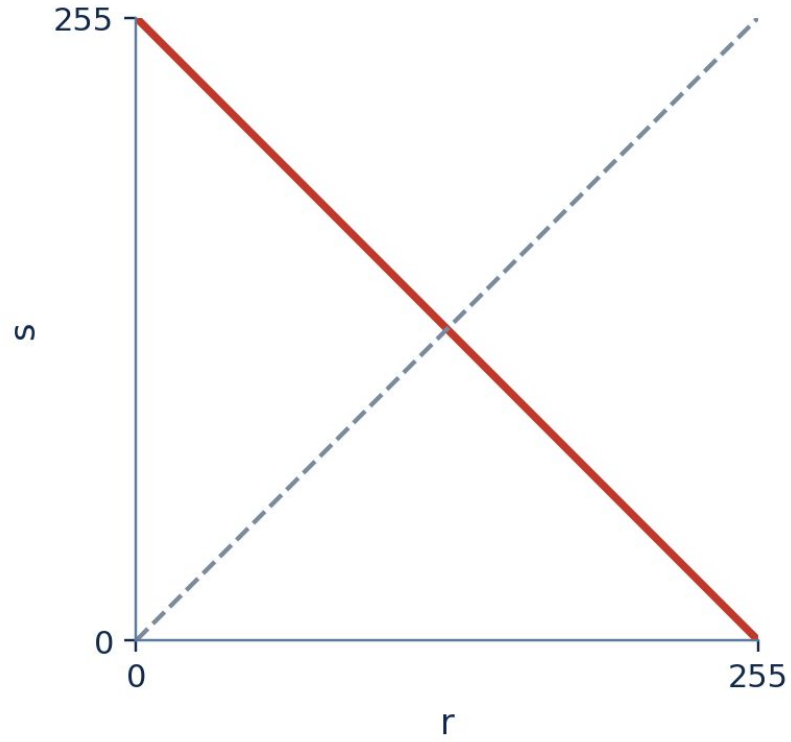
Original
(dark, low-lying detail)



Negative
$s = (L - 1) - r$

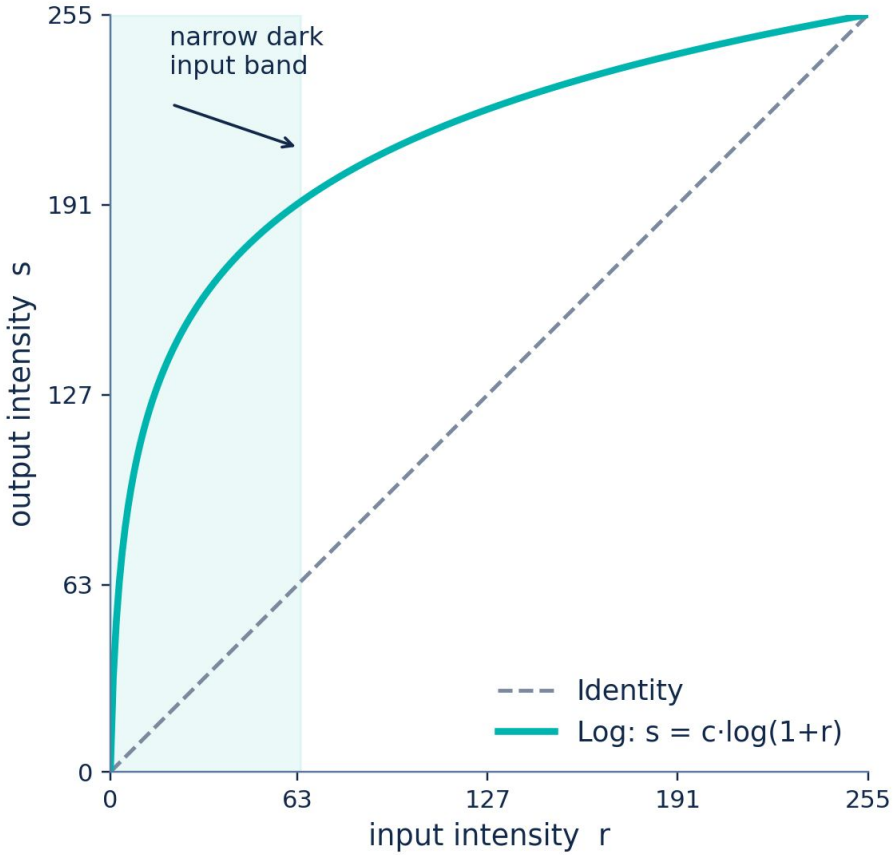


Transformation




s = (L−1) − r. Some viewers find embedded detail in dark regions easier to judge in the negative — a classic use in mammography.

# The Log Transformation



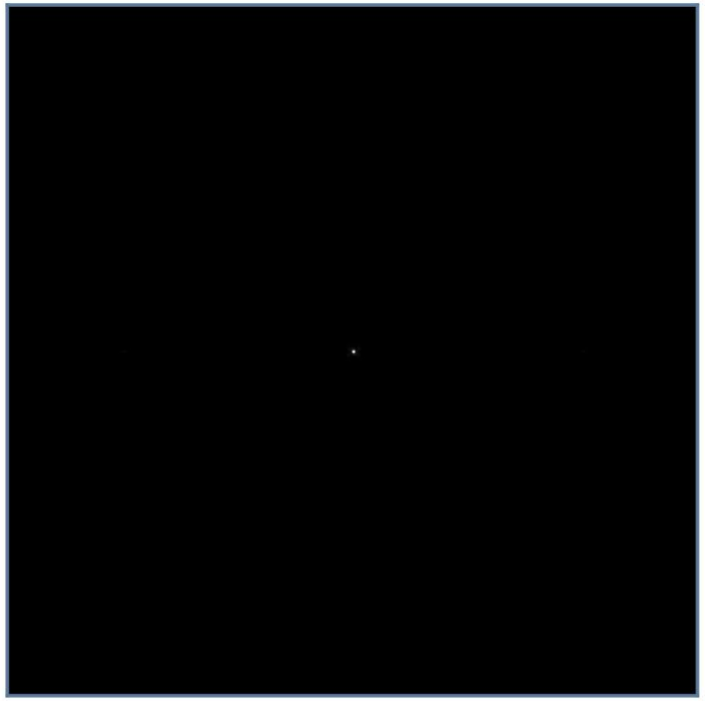

$$s = c \cdot \log(1 + r)$$

Maps a narrow band of low input intensities to a wide band of output values — and compresses high intensities into a narrow output band.

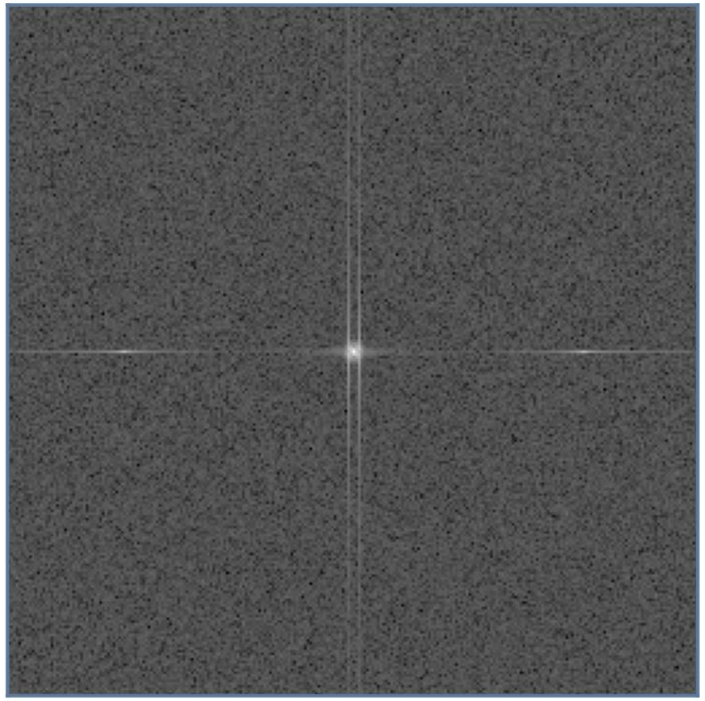
*The opposite (inverse-log / exponential) does the reverse: it expands bright values and compresses dark ones.*

# Log Transform in Practice: Compressing Dynamic Range

Spectrum, linear display
$|F(u,v)|$ scaled to [0,255]

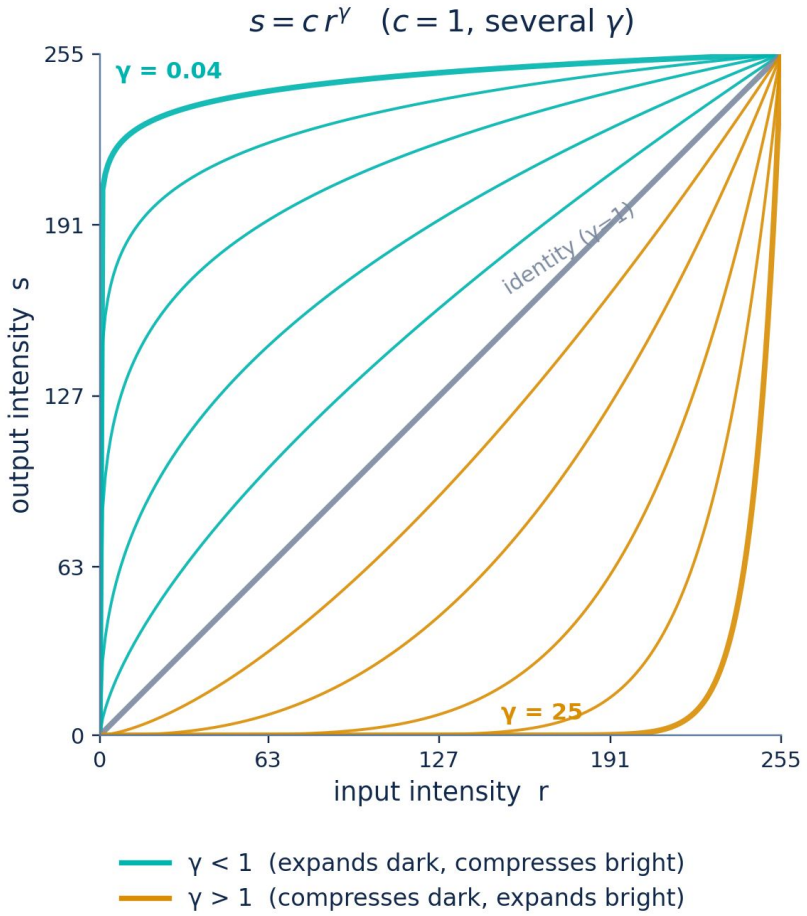


Spectrum, log display
$\log(1 + |F(u,v)|)$



A Fourier spectrum can span values from 0 to over $10^6$ — far beyond an 8-bit display. A log transform compresses this range so detail becomes visible.

# The Power-Law (Gamma) Transformation

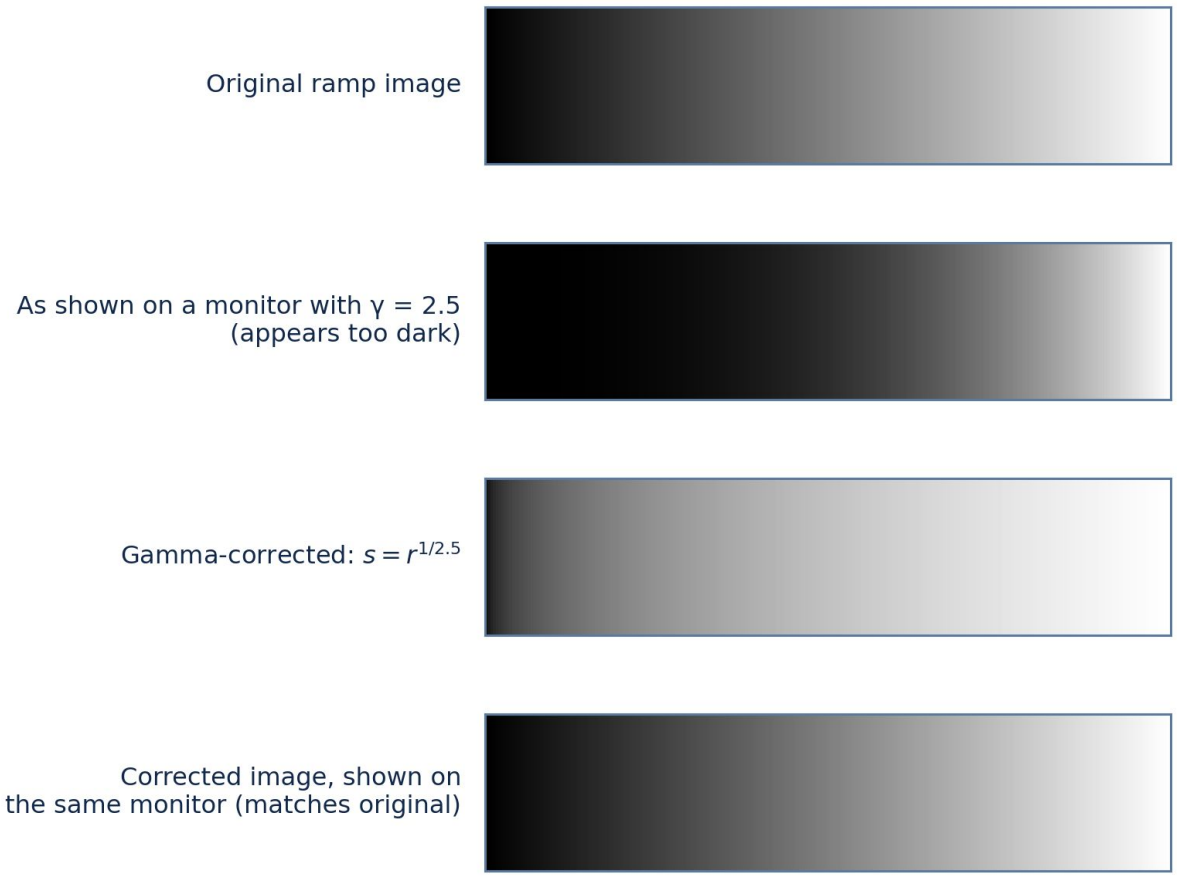



## s = c · r^γ

The single most versatile basic transformation — one parameter, γ, spans everything from the log-like curves to their opposite.

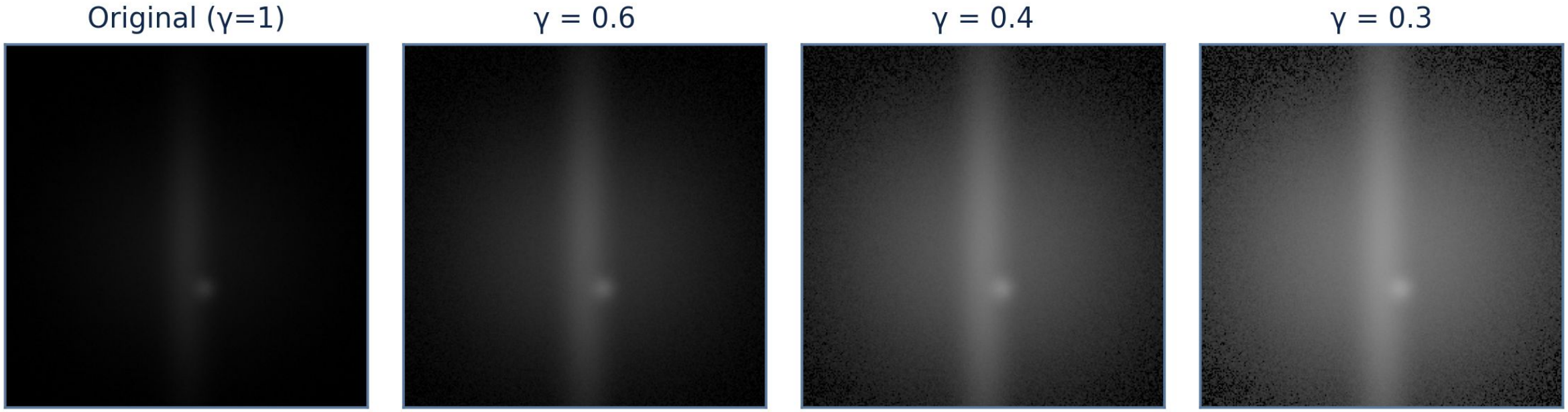
*γ = 1 reduces to the identity — no change at all.*

# Gamma Correction



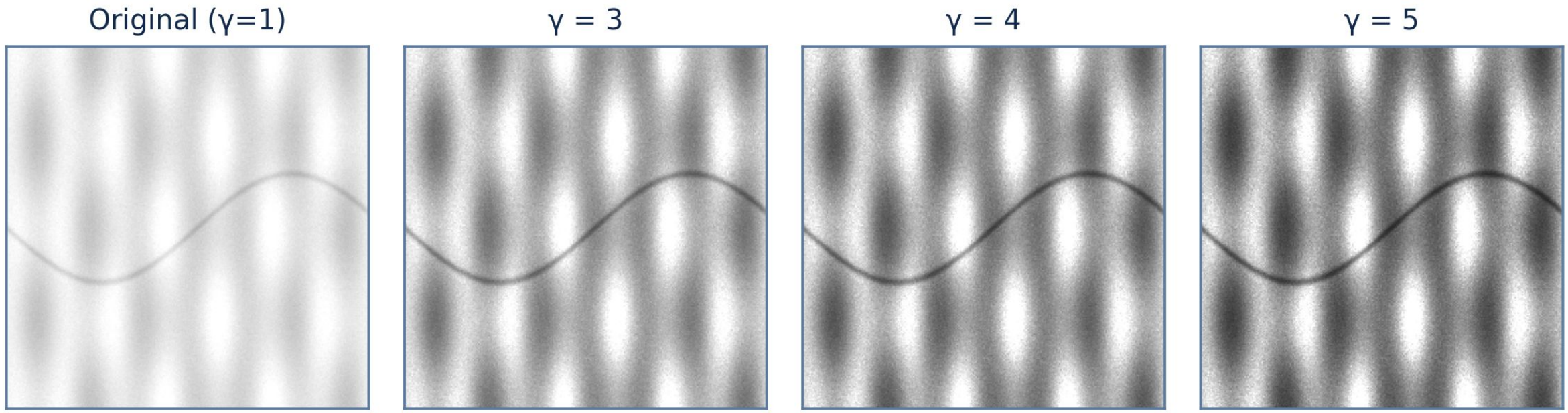

Display devices respond to input by a power law of their own (CRTs: γ ≈ 1.8–2.5). Pre-compensating with s = r^(1/γ) before display cancels the distortion.

# γ < 1: Expanding Dark Detail




A predominantly dark image, processed with γ = 0.6, 0.4, 0.3. More detail emerges as γ falls — until the background starts to wash out and noise is amplified.

# γ > 1: Compressing Washed-Out Detail




A washed-out, low-contrast image, processed with γ = 3, 4, 5. Larger γ increases contrast — useful up to the point where dark regions start to lose detail.

# Piecewise-Linear Transformations

Built from straight-line segments rather than a single formula — more work to specify, but arbitrarily flexible.

## Contrast Stretching

Expands a narrow range of intensities to span the full [0, L−1] scale.

## Intensity-Level Slicing

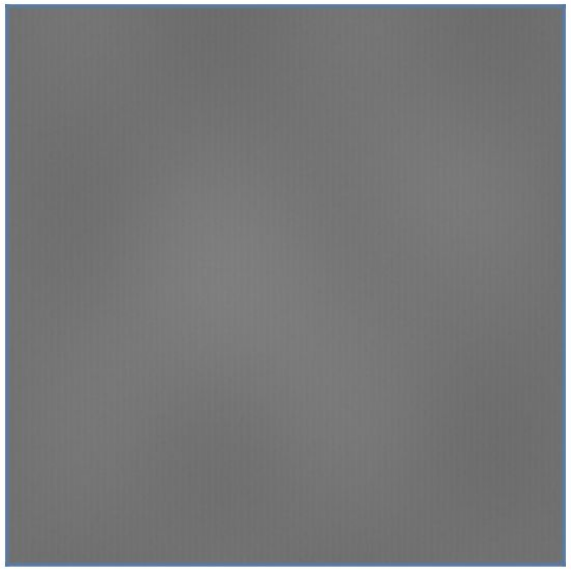
Highlights one specific band of intensities, with or without keeping the background.

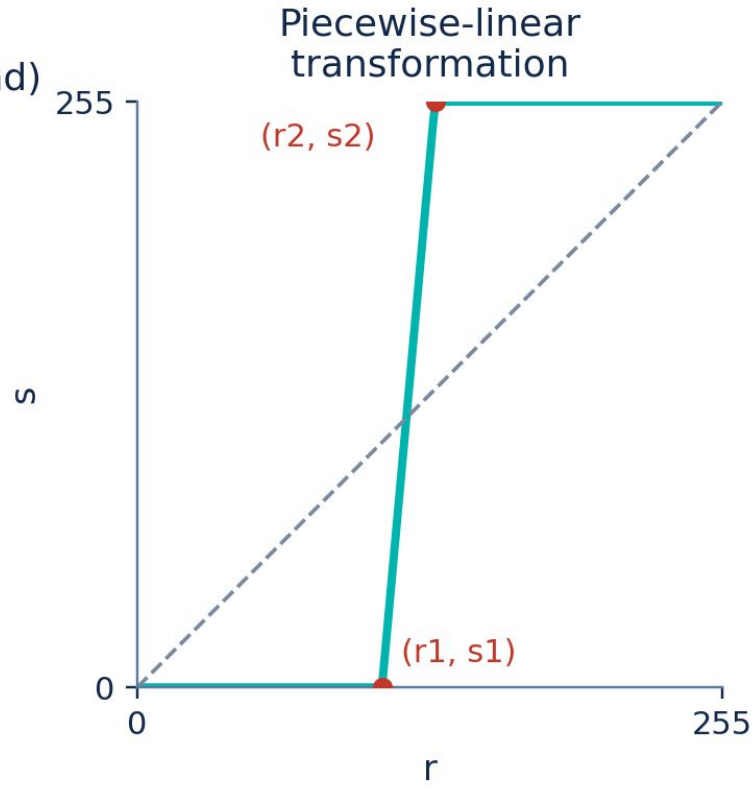
## Bit-Plane Slicing

Decomposes an image into 8 binary images, one per bit of significance.

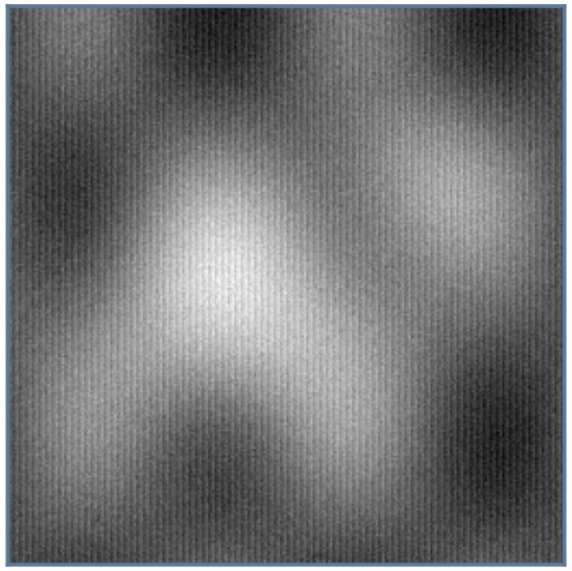
# Contrast Stretching

Low-contrast original
(intensities confined to a narrow band)





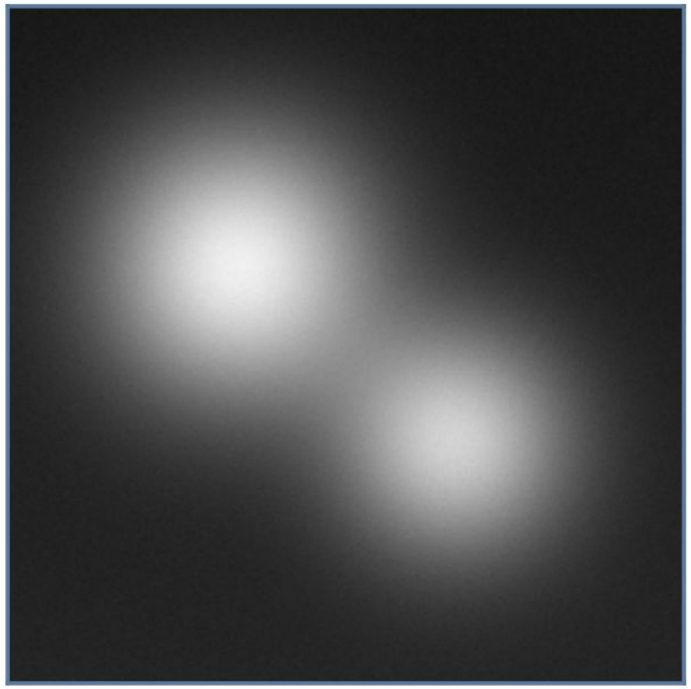

Contrast-stretched
(spans full [0, 255])



Anchor points (r1, s1) and (r2, s2) fix the line's shape. Setting them to (rmin, 0) and (rmax, L−1) stretches the image to the full intensity range.

# Intensity-Level Slicing

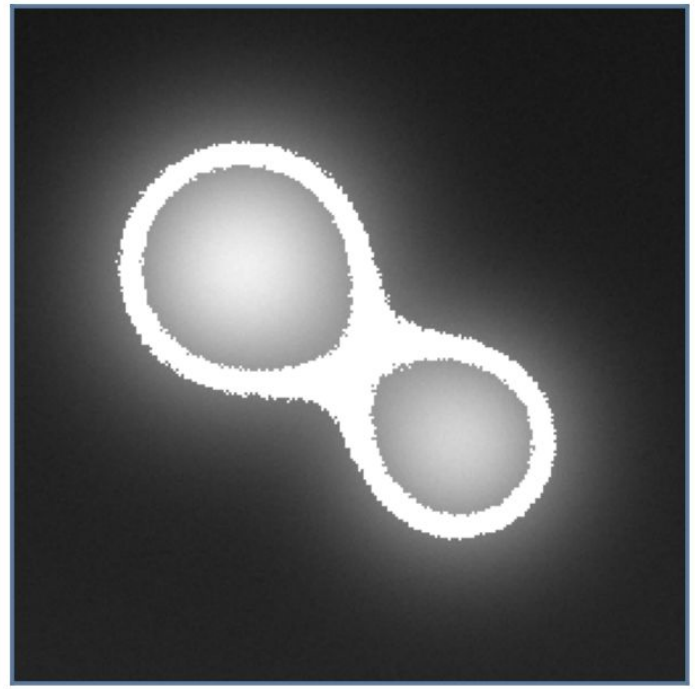
Original



Slicing: band [120,150] → white, all else → black



Slicing: band [120,150] → white, background intensities kept



Highlighting a chosen band [A, B]: either flatten everything else to black (binary result), or keep the background intact for context.

# Bit-Plane Slicing: the Idea

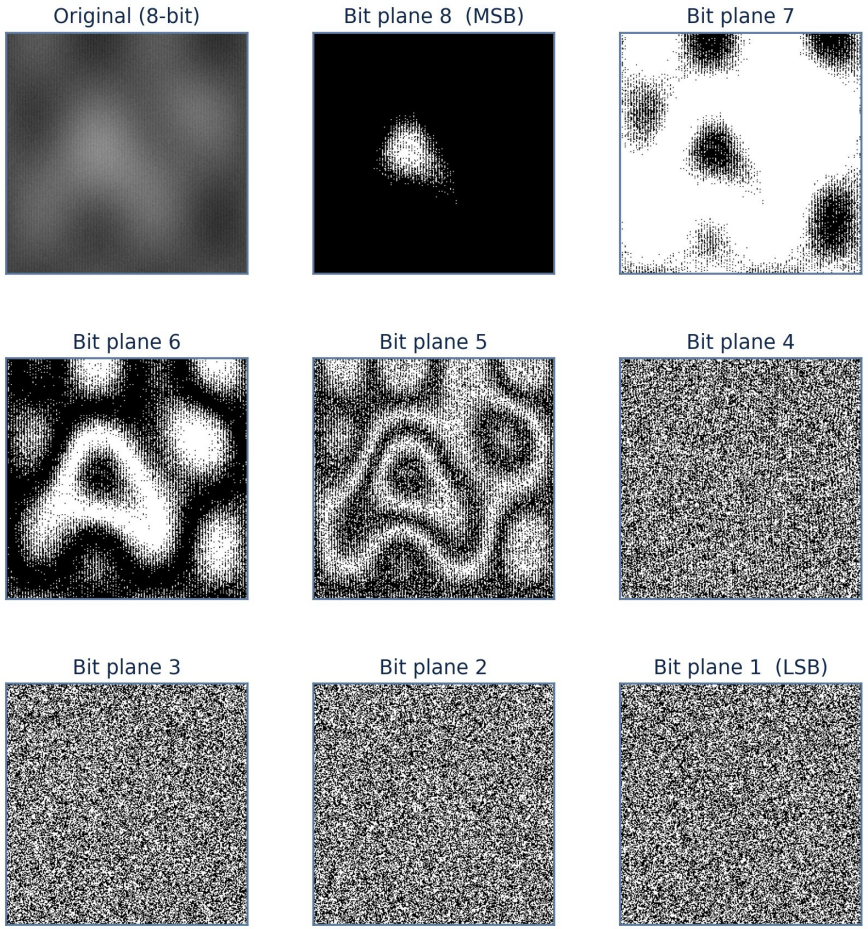
An 8-bit pixel is a byte. Each of its 8 bits, taken across every pixel, forms its own binary image — a bit plane.




*Example pixel value 194 = $11000010_2$*

**8 separate binary images — one per bit of significance**

# Bit-Plane Slicing: an Example




### Planes 8, 7, 6, 5

Carry almost all visually-significant structure.

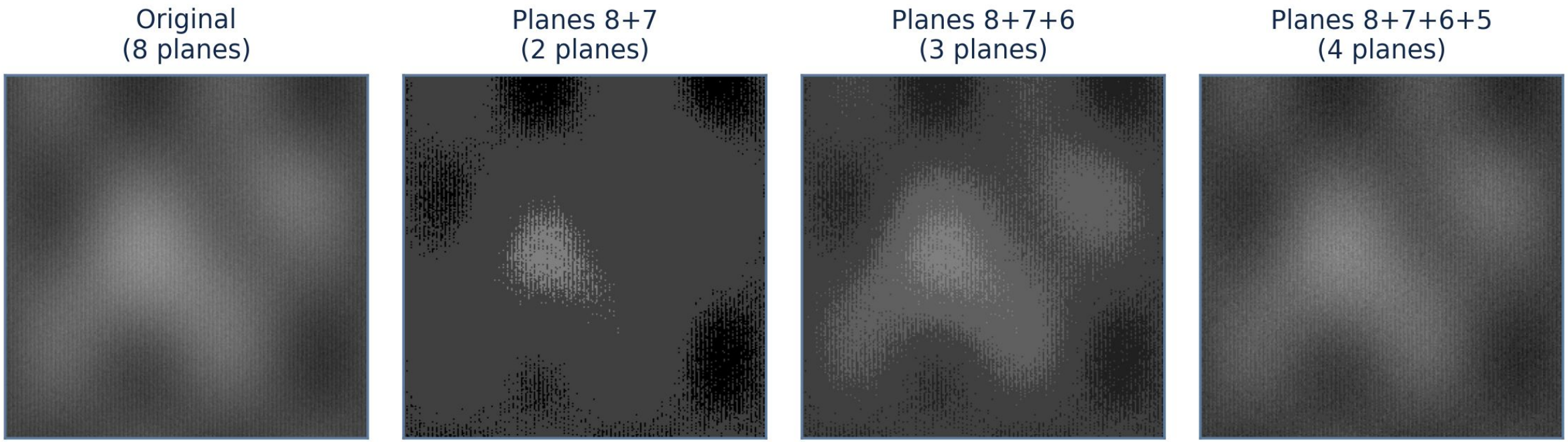
### Planes 4 through 1

Look increasingly like random noise — fine detail only.

### Bit-for-bit exact

Summing all 8 weighted planes reproduces the original image exactly — no information is lost.

# Reconstruction from Bit Planes




Summing weighted planes 2^(n−1)·plane(n) rebuilds the image. Two planes show visible false contouring; four are nearly indistinguishable from the original — at half the storage.

# Summary: Basic Intensity Transformations

| Transformation | Formula | Effect |
|---|---|---|
| **Negative** | `s = L−1−r` | Reverses the intensity scale |
| **Log** | `s = c·log(1+r)` | Expands dark, compresses bright |
| **Power-law, γ<1** | `s = c·r^γ` | Expands dark, compresses bright (tunable) |
| **Power-law, γ>1** | `s = c·r^γ` | Compresses dark, expands bright (tunable) |
| **Contrast stretching** | `piecewise-linear` | Spreads a narrow range to full scale |
| **Intensity-level slicing** | `piecewise-linear` | Highlights one chosen intensity band |
| **Bit-plane slicing** | `bitwise decomposition` | Splits the image into 8 binary planes |

NEXT CLASS

# Histogram Processing

*Equalization, matching, and statistics-based enhancement — building directly on today's point-processing model s = T(r).*

## Questions?